



**Khwaja Yunus Ali University**

## ***Lab Report -06***

**Name of the Department:** Computer Science and Engineering

**Course Code:** CSE 0713-1105

**Course Title:** Electrical Circuit Lab

**Experiment No.:** 08

**Name of the Experiment :** Measurement of power and power factor correction.

**Date of Experiment :** 23-03-2025

**Date of Submission :** 06-04-2025

.....  
**Instructor Signature & Date**

**Submitted by –**

**Name: Md Sojib Ahmed**  
ID : 06224205101006  
Batch No: 18<sup>th</sup>  
Semester: 1<sup>st</sup> Year 2<sup>nd</sup> Semester  
Khwaja Yunus Ali University

**Submitted to –**

**Name: Sakil Ahammed**  
Lecturer  
Department of EEE  
Khwaja Yunus Ali University

**No of Experiment:** 08

**Name of the Experiment:** Measurement of power and power factor correction.

**Objective:** To measure power and power factor using digital simulator.

### Theory:

Power factor ( PF ) is the ratio of real power to apparent power (  $PF = P/S = \cos \theta$  ), indicating efficiency. It is measured using a wattmeter for real power, voltmeters, and ammeters for apparent power, or directly via digital power analyzers. A low PF (closer to 0) signifies poor utilization, while unity PF (1) indicates ideal efficiency. Corrections are applied to minimize reactive power and improve PF.

### Circuit Diagram:

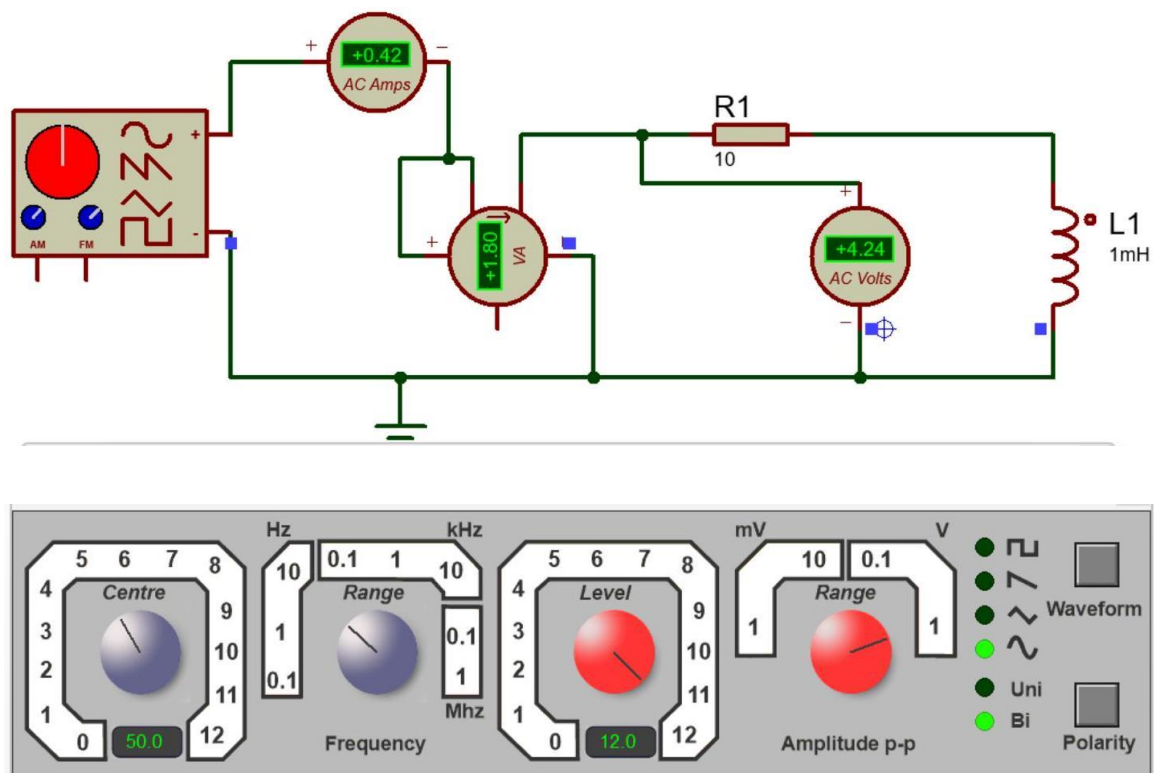


Fig : Circuit diagram of power and power factor

**List of apparatus:**

1. Software: Proteus 8 professional
2. Resistor
3. Signal generator
4. Grounding
5. Wires
6. AC Voltmeter
7. Wattmeter
8. AC Ammeter
9. Inductor

**Procedure:**

1. I placed the AC Power Supply and connected the Inductive Load (L1) in series with the ammeter (R1).
2. I connected the Voltmeter (V4) across the supply to measure input voltage.
3. I inserted the Wattmeter between the supply and load to measure real power (P).
4. Added Capacitors in series with the load for power factor correction.

**Calculation:**

Given that,

$$R = 10\Omega$$

$$f = 50 \text{ Hz}$$

$$L = 1\text{H}$$

$$V_m = 6 \text{ V}$$

First case (without capacitor),

From simulation,

$$I = 0.43 \text{ A}$$

$$P = 1.8 \text{ W}$$

$$V = 4.26 \text{ V}$$

We know,

$$\text{Apparent power, } S = VI$$

$$= 4.26 \times 0.43$$

$$= 1.8318 \text{ W}$$

So, Power Factor, P.F =  $\frac{P}{S}$

$$= \frac{1.8}{1.8318}$$

$$= 0.9826$$

Again, we know,

$$C = \frac{1}{4\pi^2 f L}$$

Capacitance,

$$= \frac{1}{4\pi^2 \times 50 \times 1 \times 10^{-3}}$$

$$= 0.5066 \text{ F}$$

Second case (with capacitor),

From simulation,

$$I = 0.42 \text{ A}$$

$$P = 1.80 \text{ W}$$

$$V = 4.24 \text{ V}$$

We know,

Apparent power,  $S = VI$

$$= 4.24 \times 0.42$$

$$= 1.7808 \text{ W}$$

So, Power Factor, P.F =  $\frac{P}{S}$

$$= \frac{1.80}{1.78}$$

$$= 1.01$$

Here,

$$X_L = X_C$$

So, the only obstacle remains is Resistance ( R ).

**Table:**

$$R = 10\Omega$$

$L = 1\text{mH}$

$C = 0.5066\text{ F}$

| Serial no. | Cases             | Voltage(V)<br>V | Current(I)<br>A | Real Power(P)<br>W | Apparent Power(S)<br>W | P.F. |
|------------|-------------------|-----------------|-----------------|--------------------|------------------------|------|
| 1          | without capacitor | 4.26            | 0.43            | 1.8                | 1.8318                 | 0.98 |
| 2          | with capacitor    | 4.24            | 0.42            | 1.7808             | 1.80                   | 1.01 |

### **Result and Discussion:**

When capacitor is added in the circuit, the value of the power factor become 1.01 which is nearly 1 . That means the maximum power is being delivered in the circuit.